

License Server Deployment on Railway with HTTPS

Public vendor license server — serial generation, client dashboard, remote revoke, push webhooks, and offline alerts

DOCUMENT PURPOSE

Deploy `license-generator/` as a secure HTTPS license server on Railway

PLATFORM

Railway.app

PROTOCOL

HTTPS (automatic TLS)

REPOSITORY PATH

`HMS_JS/license-generator`

AUDIENCE

Vendor / implementation team

Version

1.0 — June 2026

Related scripts

`license-generator/server.js`

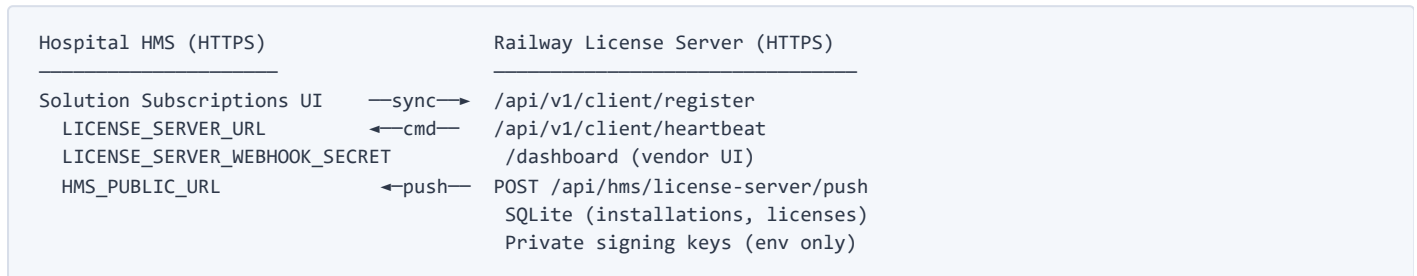
Table of contents

1	Overview & architecture	7	HTTPS & custom domain
2	Prerequisites	8	Verification & smoke tests
3	Generate licensing keys	9	Connect hospital clients
4	Create Railway project	10	Operations & security
5	Persistent SQLite volume	11	Troubleshooting
6	Environment variables		

1. Overview & architecture

The ZAIZENS license server is a Node.js Express application in `license-generator/`. It depends on shared crypto modules in the parent repo:

- `lib/hmsLicenseCrypto.js` — RSA request decryption, Ed25519 serial signing
- `lib/hmsLicenseCatalog.js` — sellable solution modules



Never deploy private keys to hospital servers. Hospitals receive public keys only (`LICENSE_RSA_PUBLIC_KEY_PEM` , `LICENSE_ED25519_PUBLIC_KEY_PEM`).

2. Prerequisites

Item	Requirement
GitHub account	Repository containing full <code>HMS_JS</code> project
Railway account	railway.app — Hobby or Pro plan recommended for volumes
Node.js locally	18+ for key generation
SMTP (optional)	For offline client email alerts
Custom domain (optional)	e.g. <code>licenses.zaizens.com</code>

3. Generate licensing keys (local, once)

1 Install dependencies

```
cd license-generator
npm install
npm run generate-keys
```

Keys are written to `license-generator/keys/`.

2 Configure license server private keys — copy into Railway variables (Section 6), never commit to Git.

3 Configure each hospital — copy **public** keys to hospital `.env` or run `node scripts/setup-hms-license-env.js`.

- 4 **Generate webhook secret** — use a long random string (32+ chars). Same value on license server and every hospital for instant push.

4. Create Railway project

- 1 Push `HMS_JS` to GitHub if not already done.
- 2 Railway → **New Project** → **Deploy from GitHub repo** → select the repository.
- 3 Open the new service → **Settings**:
 - **Root Directory**: leave empty (repo root) — required because `server.js` imports `../../lib/`
 - **Build Command**: `cd license-generator && npm install`
 - **Start Command**: `cd license-generator && npm start`
- 4 Railway sets `PORT` automatically. The server listens on `process.env.PORT`.
- 5 Wait for first deploy. Check **Deploy Logs** for: `ZAIZENS License Server listening`.

Monorepo note: Do not set Root Directory to `license-generator` alone — the build will fail because parent `lib/` files are outside that folder.

5. Persistent SQLite volume

Client registrations, license records, and remote commands are stored in SQLite at `license-generator/data/license-server.db`. Without a volume, data is lost on redeploy.

- 1 Service → **Volumes** → **Add Volume**.
- 2 Mount path: `/app/license-generator/data` (confirm your Railway root path in deploy logs — often `/app`).
- 3 Add variable: `LICENSE_SERVER_DB_PATH=/app/license-generator/data/license-server.db`
- 4 Redeploy once after attaching the volume.

6. Environment variables (Railway → Variables)

Variable	Required	Description
<code>LICENSE_RSA_PRIVATE_KEY_PEM</code>	Yes	Decrypts hospital request codes

Variable	Required	Description
<code>LICENSE_ED25519_PRIVATE_KEY_PEM</code>	Yes	Signs serial numbers
<code>LICENSE_GENERATOR_ADMIN_TOKEN</code>	Strongly recommended	Protects dashboard & admin API (<code>X-Admin-Token</code>)
<code>LICENSE_SERVER_WEBHOOK_SECRET</code>	For instant push	Shared with all hospitals
<code>LICENSE_SERVER_ALERT_EMAIL</code>	Optional	Offline alert recipients (comma-separated)
<code>LICENSE_SERVER_OFFLINE_MINUTES</code>	Optional	Default <code>45</code>
<code>LICENSE_SMTP_HOST</code> , <code>LICENSE_SMTP_FROM</code> , etc.	For email alerts	See <code>license-generator/.env.example</code>
<code>LICENSE_SERVER_DB_PATH</code>	With volume	Absolute path on volume mount

PEM formatting in Railway

Paste multi-line PEM directly, or single-line with `\n` between lines. The application normalizes both formats.

7. HTTPS & custom domain

7.1 Railway default domain (automatic HTTPS)

- 1 Settings → **Networking** → **Generate Domain**.
- 2 Railway provides e.g. `https://zaizens-license-production.up.railway.app` — TLS is automatic.
- 3 Use this URL as `LICENSE_SERVER_URL` on hospitals (no trailing slash).

7.2 Custom domain

- 1 Networking → **Add Custom Domain** → e.g. `licenses.zaizens.com` .
- 2 Create DNS **CNAME** pointing to Railway's target hostname.
- 3 Wait for certificate provisioning (Let's Encrypt, automatic).
- 4 Update hospital `LICENSE_SERVER_URL` to the custom domain.

8. Verification & smoke tests

Test	URL / command	Expected
Health	<code>GET /api/health</code>	<code>ok: true</code> , <code>keysConfigured: true</code>

Test	URL / command	Expected
Dashboard	<code>/dashboard</code>	Login with admin token; client list loads
Generator	<code>/</code>	Serial generator UI loads
From hospital	Subscriptions → Sync now	Installation appears in dashboard

```
curl https://YOUR-DOMAIN/api/health
```

9. Connect hospital clients

On each Windows hospital server `.env`:

```
LICENSE_SERVER_URL=https://licenses.zaizens.com
LICENSE_SERVER_SYNC_INTERVAL_MS=900000
LICENSE_SERVER_WEBHOOK_SECRET=same-secret-as-railway
HMS_PUBLIC_URL=https://hms.client-hospital.org
LICENSE_RSA_PUBLIC_KEY_PEM="..."
LICENSE_ED25519_PUBLIC_KEY_PEM="..."
HMS_LICENSE_DEPLOYMENT=1
```

Restart Node HMS. Open `/hms-admin/subscriptions` → confirm license server panel → click **Sync now**.

10. Operations & security

- ☐ Rotate `LICENSE_GENERATOR_ADMIN_TOKEN` periodically
- ☐ Backup SQLite volume / export `license-server.db` weekly
- ☐ Monitor Railway deploy logs for push webhook failures
- ☐ Revoke compromised installations from dashboard immediately
- ☐ Keep Railway project access limited to vendor staff
- ☐ Document which hospitals share the same webhook secret

11. Troubleshooting

Symptom	Cause / fix
<code>keysConfigured: false</code>	Private PEM variables missing or malformed
Clients disappear after redeploy	No volume on <code>license-generator/data</code>
Cannot find module '../../lib/hmsLicenseCrypto'	Root Directory must be repo root, not subfolder only
Push webhook fails	Hospital <code>HMS_PUBLIC_URL</code> must be public HTTPS; secret must match
better-sqlite3 build error	Ensure Node 18+; redeploy; check Railway build logs
No offline emails	Set <code>LICENSE_SERVER_ALERT_EMAIL</code> and SMTP variables